

Grade 8
Unit 2
Lesson 7 Practice

Name Answer Key
Date _____
Period _____

1. Identify an irrational number between -4 and -5 .

Answers will vary.

Possible answers include:

$$-\sqrt{25} < x < -\sqrt{16}, \sqrt[3]{-125} < x < \sqrt[3]{-64}$$

2. Victor thinks that $-\frac{28}{6}$ is greater than $\sqrt[3]{-100}$. Sean thinks that $\sqrt[3]{-100}$ is greater. Who is correct? Explain.

$$\sqrt[3]{-100} \approx -4.64$$

$$\frac{-28}{6} = -4.\bar{6}$$

$$\frac{17}{4} = 4.25$$

Sean is correct, $\frac{17}{4}$ is positive while $\frac{-28}{6}$ and $\sqrt[3]{-100}$ are both negative.

Compare each pair of expressions using $<$, $>$, or $=$.

3. $-2\pi + 10$ ☒ $>$ $\sqrt[3]{-56}$

4. $\sqrt{300}$ ☒ $<$ $\frac{196}{11}$

5. $-\sqrt{\frac{162}{2}}$ ☒ $>$ $-9.\overline{09}$

6. $3\sqrt{100}$ ☒ $<$ π^3

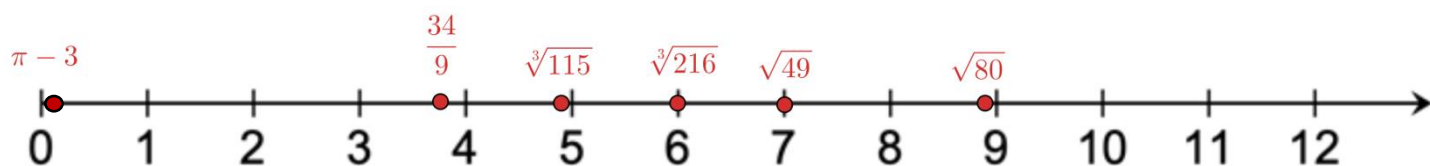
7. Order the following values from least to greatest.

$$\pi - 3, \quad \sqrt{80}, \quad \sqrt[3]{216}, \quad \frac{34}{9}, \quad \sqrt{49}, \quad \sqrt[3]{115}$$

$$\approx 0.1416 \quad \approx 8.94 \quad = 6 \quad = 3.\bar{7} \quad = 7 \quad \approx 4.86$$

$$\pi - 3, \frac{34}{9}, \sqrt[3]{115}, \sqrt[3]{216}, \sqrt{49}, \sqrt{80}$$

8. Plot and label each value from the previous question on the number line.



For questions 9-10, order each set of numbers from least to greatest.

9. $\sqrt{400}, \quad \frac{78}{4}, \quad 2\pi^2$

$$\frac{78}{4}, \quad 2\pi^2, \quad \sqrt{400}$$

10. $\sqrt[3]{-126}, \quad -\sqrt{25}, \quad -5.0\bar{1}$

$$\sqrt[3]{-126}, \quad -5.0\bar{1}, \quad -\sqrt{25}$$